

This manual is for interfacing RP LiDAR A2M8 with Jetson Nano

Prerequisites:

1. Properly flashed SD card with the most recent jetpack version
2. Ethernet or WiFi connection
3. Admin permission on the Jetson's Ubuntu GUI
4. Keyboard, mouse, monitor connected to the Jetson
5. RP LiDAR A2M8 connected to the jetson by USB-A

Step-1: Installing Build tools

Run:

```
sudo apt update
```

```
sudo apt install git g++ build-essential
```

Step-2: Cloning the SDK

Run:

```
cd ~
```

```
git clone https://github.com/Slamtec/rplidar_sdk.git
```

```
cd rplidar_sdk
```

Step3: Compiling the Provided Makefile

Run:

```
make
```

Verify by running

```
cd ~/rplidar/output/Linux/Release
```

Run a demo

```
./ultra_simple --channel --serial /dev/ttyUSB0 115200
```

-----End of bare minimum-----

Setting Up Catkin space

Step-1:

Run:

```
mkdir -p ~/catkin_ws/src
```

then

```
cd ~/catkin_ws/src
```

```
git clone https://github.com/Slamtec/rplidar_ros.git
```

then

```
cd ~/catkin_ws
```

```
catkin_make
```

then

nano ~/.bashrc

go to the bottom of the file and add these two lines

```
source /opt/ros/melodic/setup.bash
```

```
source ~/catkin_ws/devel/setup.bash
```

press ctrl+o then enter, and lastly ctrl+x

then

```
source ~/.bashrc
```

Verify by running

```
roscd rplidar_ros
```

this should drop in the right folder ~/catkin_ws/src/rplidar_ros

Step-2: Create a udev rule for automatic serial permissions

Run:

```
sudo nano /etc/udev/rules.d/99-rplidar.rules
```

paste this in the file

```
SUBSYSTEM=="tty", ATTRS{idVendor}=="10c4", ATTRS{idProduct}=="ea60",  
MODE="0666", GROUP="dialout"
```

press ctrl+o then enter, and lastly ctrl+x

then run

```
sudo udevadm control --reload-rules
```

```
sudo udevadm trigger
```

run:

```
source ~/catkin_ws/devel/setup.bash
```

and finally:

```
roslaunch rplidar_ros rplidar_a2m8.launch
```

this should return

- ☐ SDK Version: 2.x.x
- ☐ Start motor...
- ☐ health status : Good
- ☐ RPLIDAR running on device /dev/ttyUSB0 at 115200